SOFTWARE REQUIREMENTS SPECIFICATION

Agricultural Wages Analytics Platform (AWAP) — CSV Data Pipeline

DOCUMENT ID	TARGET DATASET	COMPLIANCE	STATUS
SRS-AWAP-2026-V1	Agricultural Wages_Sample_Data.csv	IEEE Std 830-1998 / ISO 29148	Approved Baseline

1. EXECUTIVE SUMMARY & SCOPE

This Software Requirements Specification (SRS) defines the functional, structural, and quality requirements for the **Agricultural Wages Analytics Platform (AWAP)**. The system automates ingestion, schema enforcement, spatial-temporal aggregations, and disparity modeling on rural agricultural wage records ingested via `Agricultural Wages_Sample_Data.csv`.

> **CORE SYSTEM MISSION**
>
> The AWAP ingests state, district, and activity-level rural wage metrics across male and female workforces, generating wage disparity indicators, gender pay gap analysis, seasonal inflation adjustments, and automated statutory minimum wage compliance alerts.

2. SOURCE DATASET SCHEMA DEFINITION

The system shall process raw CSV datasets conforming strictly to the structural and field-level specification defined below:

Field Name	Data Type	Validation Rule / Constraints	Sample Value
`State_Name`	String (50)	Mandatory. Standardized Indian State/UT string.	Andhra Pradesh
`District_Name`	String (50)	Mandatory. Standardized District name mapped to LGD codes.	Guntur
`Year`	Integer	YYYY format; range: 2000 ≤ Year ≤ 2030.	2024
`Month`	String (3)	Enum: [Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec].	Jan
`Activity_Category`	Enum	Values: [Agricultural, Non-Agricultural].	Agricultural
`Activity_Name`	String (100)	Standardized list (Ploughing, Sowing, Harvesting, General Labour).	Ploughing/Tilling
`Wage_Male_INR`	Decimal (8,2)	Range: 0.00 to 5000.00; Null allowed if work unperformed.	642.50
`Wage_Female_INR`	Decimal (8,2)	Range: 0.00 to 5000.00; Null allowed if work unperformed.	414.70
`Data_Source`	String (50)	Labour Bureau / DES / State Ag Dept source identifier.	Labour Bureau
`Record_Status`	Enum	Values: [Validated, Flagged, Under_Review, Archived].	Validated

3. SYSTEM ARCHITECTURE & DATA PIPELINE FLOW



4. SPECIFIC FUNCTIONAL REQUIREMENTS

REQ ID	Requirement Subsystem	Detailed Functional Specification
FR-ING-01	Batch CSV Import	The system shall ingest `Agricultural Wages_Sample_Data.csv` files up to 500 MB via REST API and Web Interface within 5 seconds.
FR-VAL-02	Anomalous Outlier Detection	The system shall calculate Z-scores for male and female wages grouped by `State_Name` and `Activity_Name`. Values outside 3 standard deviations shall be flagged as `Flagged_Outlier`.
FR-MTR-03	Gender Wage Gap Calculation	The system shall automatically compute the percentage disparity using the formula: Disparity (%) = [(Wage_Male_INR − Wage_Female_INR) / Wage_Male_INR] × 100.
FR-AGG-04	Multi-Dimensional Aggregations	The system shall aggregate data by Annual Average, District Rank, State Index, and Activity-Specific Median Wage across custom date ranges.
FR-CMP-05	Statutory Compliance Monitoring	The system shall evaluate recorded wages against official minimum wage notifications published by the Chief Labour Commissioner (Central/State). Non-compliant records must trigger high-priority alerts.
FR-EXP-06	Export & Reporting Engine	The system shall generate filtered report outputs in PDF, XLSX, and GeoJSON formats containing spatial-temporal visualizations and statistical summary tables.
FR-SEC-07	Role-Based Access Control	The system shall enforce RBAC across three tiers: *Administrator* (Full Access/Upload), *Analyst* (Query/Export), and *Public Reader* (Read-only Aggregates).

VALIDATION RULE EXAMPLE

If `Wage_Male_INR` < 0 or `Wage_Female_INR` < 0, the row is rejected immediately to Error Logs. Missing gender wage data marked as '@' or '-' in CSV inputs must be mapped to `NULL` without breaking calculation streams.

5. NON-FUNCTIONAL REQUIREMENTS (NFR)

Category	Metric / Target Standard	Implementation Specification
Performance	Query Response < 800ms	95% of analytical queries over 10 million rows must execute within 800 milliseconds using pre-aggregated materialized views.
Scalability	Horizontal Scale	System architecture shall handle concurrent ingestion of up to 50 stream pipelines without degradation of dashboard rendering speed.
Availability	99.9% Uptime	Deployment in high-availability Kubernetes cluster with automated failover and dual-region database replication.
Security	AES-256 / TLS 1.3	All persistent wage data encrypted at rest (AES-256) and in transit via TLS 1.3 protocol with OAuth 2.0 / OIDC authentication.
Integrity	Audit Logging	100% of upload events, record edits, and export requests captured in an immutable audit trail with cryptographic timestamps.

6. INTERFACE & API SPECIFICATIONS

The platform shall expose standardized REST APIs for automated data pipeline integration with external governmental and economic portals:

HTTP Method	Endpoint Path	Request Payload / Params	Response Description
POST	`/api/v1/wages/ingest`	Multipart CSV file upload	Returns processing summary, record counts, and rejected error log URLs.
GET	`/api/v1/wages/query`	`?state=AP&year=2024&activity=Sowing`	JSON dataset containing aggregated mean, median, and gender pay gap metrics.
GET	`/api/v1/wages/disparity-index`	`?state=ALL&year=2024`	State-level comparative ranking array based on calculated gender wage gaps.

7. VERIFICATION & TRACEABILITY MATRIX

QUALITY ASSURANCE STANDARDS

All requirements defined in FR-ING-01 through FR-SEC-07 shall undergo 100% automated unit testing, integration test suites, and load testing prior to release approval.